

AI Portfolio

SOW Orchestrator

Autonomous 3-Agent AI Pipeline

Discovery · Validation · SOW Drafting

Demo Mode

Claude API

OpenAI API

danvzla.github.io/ai-sow-orchestrator

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THE PROBLEM

SOW creation is one of the most manual, time-consuming tasks in professional services



2–3 days per engagement

Senior architect time spent structuring notes, scoping deliverables, estimating effort, and drafting a professional SOW — every time, from scratch.



Inconsistent quality

SOW quality depends on who wrote it. Scope gaps, missing assumptions, and vague acceptance criteria create disputes and margin erosion during delivery.



No quality gate

Notes go from meeting room to SOW with no structured review. Contradictions, unrealistic timelines, and missing dependencies make it to the client.

THREE MODES

Zero setup to live AI drafting — one click apart



No API key required

Demo Mode

Click Demo → select a scenario → click Generate SOW. Full 3-agent simulation runs in ~6 seconds. All 4 output tabs populate. Works offline.

Best for: Always ready · no setup · no network dependency



claude-haiku-4-5

Claude API

Select Claude → enter sk-ant-... key → Connect → Generate SOW. Real 3-agent pipeline calls Anthropic API with your actual discovery notes.



gpt-4o-mini

OpenAI API

Select OpenAI → enter sk-... key → Connect → Generate. Same 3 prompts, same 4 output tabs, same SOW structure. One click to switch providers.

FOUR OUTPUT TABS

Each agent's output in its own scrollable tab

01

Pipeline

Live agent animation — 3 cards with progress bars and status updates as each agent runs. See exactly what's happening in real time.

- ◆ Agent 1, 2, 3 cards with status text
- ◆ Gold progress bars animate per agent
- ◆ Spinner → check → done transitions
- ◆ Pipeline sub-status updates throughout

02

Agent 1 — Brief

The structured requirements brief Agent 1 extracted from your raw notes. Two-column layout: scope on left, risks on right.

- ◆ Engagement objective (1 sentence)
- ◆ In scope vs out of scope workstreams
- ◆ Deliverables, phases, success criteria
- ◆ Risks, assumptions, open items

03

Agent 2 — Validation

Agent 2's autonomous quality decision. Approved / Approved with warnings / Rejected — with quality score and issue list.

- ◆ Decision badge with quality score 0–100
- ◆ Validation summary paragraph
- ◆ Issues listed with fix recommendations
- ◆ Warnings passed to Agent 3 SOW risk register

04

Agent 3 — SOW

The complete 14-section Statement of Work. Fully scrollable. Print to PDF or copy to clipboard with one click.

- ◆ 14 sections with tables throughout
- ◆ Effort estimates and phase breakdown
- ◆ Risk register, milestones, commercial terms
- ◆ Print to PDF / Copy to clipboard buttons

HOW IT WORKS

Autonomous 3-agent pipeline — Agent 3 is gated by Agent 2

1

0 sec

Select mode & load notes

Choose Demo, Claude, or OpenAI.
Paste raw discovery notes or load one
of 6 pre-built scenarios. Set
engagement type, size, and industry.

2

1–15 sec

Agent 1 — Discovery Analyst

Reads raw notes. Extracts
workstreams, deliverables, phases,
assumptions, risks, and open items
into a structured JSON brief.

3

1–10 sec

Agent 2 — Scope Validator

Reviews the brief for contradictions,
gaps, and ambiguity. Returns:
approved / approved_with_warnings /
rejected + quality score.

4

5–30 sec

Agent 3 — SOW Drafter (if approved)

Only runs if Agent 2 approved. Writes
a complete 14-section SOW in HTML
format with tables, effort estimates,
and commercial terms.

🚫 Agent 3 is gated — it only runs if Agent 2 returns "approved" or "approved_with_warnings". A rejection blocks SOW drafting and shows the specific issues to fix.

Browser-native · three layers · no server required

PRESENTATION

- ◆ `index.html` — single file, zero dependencies
- ◆ Left panel: fixed sidebar (inputs, controls)
- ◆ Right panel: 4-tab output (Pipeline/Brief/Validation/SOW)
- ◆ Tab bar appears after first run
- ◆ Provider pills: Demo · Claude · OpenAI

GitHub Pages ready
Open `index.html` directly

ORCHESTRATION

- ◆ `setMode()` — switches provider, updates UI
- ◆ `runOrchestrator()` — routes demo vs API path
- ◆ `showPipeline()` — animated agent pipeline
- ◆ `switchTab()` / `enableTab()` — tab controller
- ◆ `setAgentState()` + `animateProgress()` — live animation

Pure JavaScript
No build step needed

AI + DATA

- ◆ `DEMO_ANALYSIS` — pre-built VCF scenario output
- ◆ `callClaude()` → claude-haiku-4-5
- ◆ `callOpenAI()` → gpt-4o-mini
- ◆ `repairAndParseJSON()` — 4-attempt repair chain
- ◆ 6 pre-built discovery scenarios (vcf/nsx/telco/sase/mod/noc)

Claude or OpenAI
1-click provider swap

Same 3-agent pipeline — runs from your terminal

sow_generator.py

Identical 3-agent pipeline, identical prompts, identical output — runs on Claude or OpenAI from the command line. Outputs Markdown and a styled Word document (.docx) with charcoal & gold formatting.

- ◆ Claude (claude-haiku-4-5) or OpenAI (gpt-4o-mini) via --provider flag
- ◆ Rich terminal output with spinners and progress (degrades gracefully)
- ◆ Word .docx export — charcoal & gold theme, tables, effort estimates
- ◆ 4-attempt JSON repair chain — same resilience as the web tool
- ◆ 6 engagement types · 4 sizes · 9 industries · --interactive mode

Terminal

Install

```
pip install anthropic rich python-docx
```

Claude (default)

```
python sow_generator.py notes.txt
```

OpenAI

```
python sow_generator.py notes.txt  
--provider openai
```

Interactive

```
python sow_generator.py --interactive
```

Full options

```
python sow_generator.py notes.txt  
--type "Telco NFV" --size "Large"
```

📄 Markdown output

SOW_vcf_20260617.md

📄 Word document

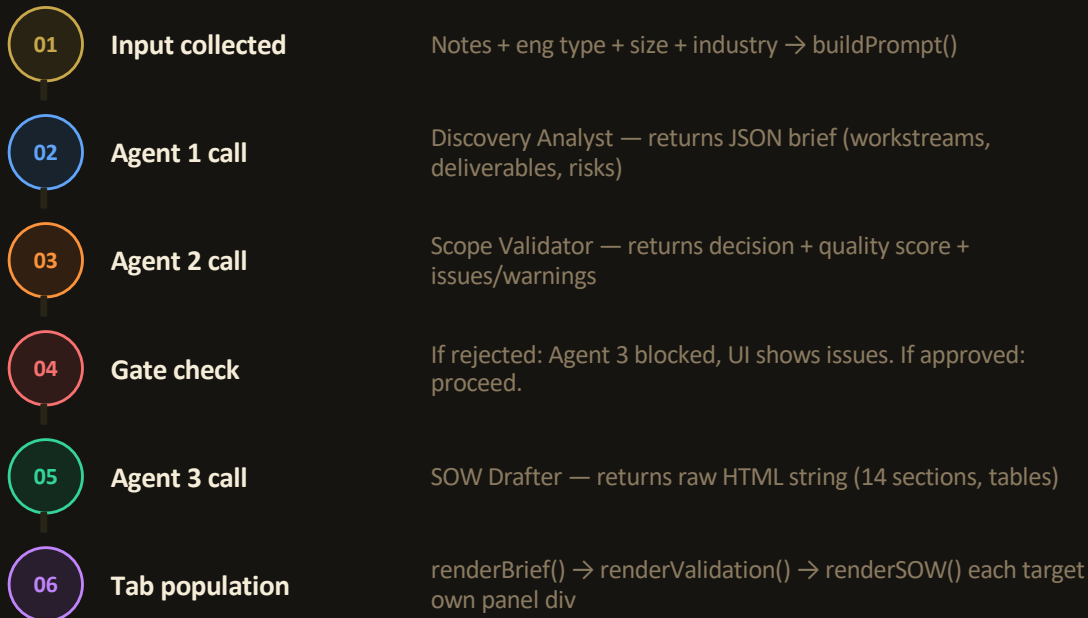
SOW_vcf_20260617.docx

🔑 API key via env

ANTHROPIC_API_KEY or OPENAI_API_KEY

Data flow · JSON repair · provider abstraction

INPUT → AGENT 1 → AGENT 2 → AGENT 3 → OUTPUT



JSON Repair Chain

LLMs sometimes return malformed JSON. All 3 agents run through 4-attempt recovery:

- 1 Direct `JSON.parse()`
- 2 Strip control chars
- 3 Escape bare newlines
- 4 Close open brackets

Attempt 4 handles token-limit truncation — closes unclosed { and [brackets

SAMPLE OUTPUT

VCF Private Cloud · Financial Services · 14-week engagement

⚠ Agent 2 — Approved with Warnings · Quality score: 88/100

Scope is well-structured with clear objectives and realistic phasing. Two warnings added to the SOW risk register.

Agent 1 — Requirements Brief

Workstreams	5 in scope (VCF, NSX, vSAN, VM migration, ops)
Deliverables	5 (reference architecture, NSX policy, runbooks, DR report, handover)
Duration	14 weeks · 3 phases
Team	Principal Architect, Network Engineer, Migration Specialist, TPM

SOW Section

4. Deliverables

VCF Reference Architecture · NSX Policy · VM Migration Runbooks (×6) · DR Report · Ops Handover Package | Total: 236 hrs

SOW Section

5. Work Breakdown

Phase 1: Foundation (4wk, 152hrs) · Phase 2: NSX & DR (5wk, 128hrs) · Phase 3: Migration & Handover (5wk, 128hrs) · TOTAL: 408 hrs

SOW Section

10. Risk Register

Hardware certification gap · Change management delays · Storage team resistance · NSX skills gap · Overrun perception risk

SOW Section

13. Commercial

Fixed fee · 25% at kickoff · 25% at M1 · 25% at M3 · 25% at M5 · 30-day hypercare included

KEY BENEFITS

What this demonstrates — and what it replaces

<6s

Demo Mode

Full 3-agent pipeline, no API key

4

Output tabs

Pipeline·Brief·Validation·SOW

3

Providers

Demo · Claude API · OpenAI API

14

SOW sections

Deliverables, WBS, risk register & more

6

Scenarios

VCF · NSX · Telco · SASE · Cloud · NOC

QA

Autonomous gate

Agent 2 blocks bad briefs from Agent 3

6 PRE-BUILT SCENARIOS

Load any scenario in Demo Mode — no typing needed

01

VCF Private Cloud Deployment

Tools: VMware NSX · vDefend ATP · SD-WAN · SASE · Compliance: NIST CSF · SOC 2

Best for:

VMware / Broadcom · Cisco accounts

02

NSX Security Hardening

Tools: NSX DFW · vDefend ATP · Zero Trust · Compliance: HIPAA

Best for:

Healthcare · compliance-driven

03

Telco NFV Transformation

Tools: OpenStack · VNF onboarding · SD-WAN · Compliance: ETSI NFV

Best for:

Nokia · Ericsson · AT&T · T-Mobile

04

SASE / SD-WAN Migration

Tools: VMware SASE · Palo Alto Prisma · Fortinet · Compliance: ISO 27001

Best for:

Manufacturing · retail SI accounts

05

Cloud Modernization Assessment

Tools: AWS · EKS · FinOps · Kubernetes · Compliance: —

Best for:

Technology · SaaS · FinOps engagements

06

NOC Automation Program

Tools: SolarWinds · Nagios · ServiceNow · AI triage · Compliance: —

Best for:

Cable MSO · Telco operations

LIVE DEMO · NO API KEY REQUIRED

Try it right now

Live demo → danvzla.github.io/ai-sow-orchestrator

GitHub → github.com/danvzla/ai-sow-orchestrator

- ◆ Click Generate SOW → Demo Mode → full 3-agent pipeline in ~6 seconds, no API key
- ◆ 4 output tabs: Pipeline · Agent 1 Brief · Agent 2 Validation · Agent 3 SOW
- ◆ 6 pre-built scenarios: VCF · NSX · Telco · SASE · Cloud · NOC Automation
- ◆ Switch to Claude or OpenAI for live AI generation with your own key

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AI · SOW Orchestrator

